CPU EDUCATION

COMP90051 期末班

Final ȕ Ӥˀ 讲解

TUTOR：JJ 学长

时长：1 小时

unimelb

本周内容预览：

1.个人介绍

2.考试介绍

3.CPU 期末班针对方案

4.统计流派：频率和贝叶斯

5.线性回归

6.逻辑回归

7.往年相关习题练习

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CPU EDUCATION

关于CPU EDUCATION

CPU Education成立于2018年，总部在澳大利亚墨尔本，目前国内办公室在广州。经过近多年的积累与发展，公司凭借优质的教学质量和专业的研发团队力量不断壮大，成为全球最受学生欢迎的教育机构之一，在众多学生及老师中享有良好的口碑。CPU Education争做行业标杆，为不同阶段的学生提供专业，专属，专注的教学服务。

致力于为学生实现人人H1的目标，以“为学生提供极具价值的教学服务，培训各行业精英”为宗旨。秉承“教育高于一切”的经营理念，本若服务定制化、师资专业化、教学规范化的要求，打造墨尔本教育行业的“黄埔军校”。

课后，如果您有任何建议和意见，我们都非常欢迎您联系小助手分享您的想法，给予我们改进和提高的机会!
感谢您参与CPU Education的辅导课程!

课程反馈

PART 1 个人介绍

Name： CPU JJ，一个熟悉考试的小镇做题家

Master of data science 所学课程成绩均为 80+， WAM 88 分。其中 SML 这门课 89 分，擅长统计，数据库及机器学习

1 年 tutor 经验，熟悉 SML 的考点及考纲

擅长按部就班给学生总结知识点，以普通人的方式打开知识大门

擅长高效制定备考计划，高效复习知识以及科学刷题，帮助你轻松掌握考点及成功晋升小镇做题家

同学问？期末和平时班的 JJ 有什么不同？

Ans： 期末班的 JJ 备课会更有针对性，重点集中在讲解考点和得分点从平时班的注重知识的广度变为集中为考点的深度。

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PART 2 考试介绍

Hurdle，考试时长 2h，占总分值的百分之 40

考试内容：0 代码考察，考察统计机器学习算法理论和计算

试卷结构（这个学期开始首次闭卷）：

40 分简答题（有些问题考察理解点细致刁钻点，但也尽量拿）：

平均下来 3-5 分一题，大概 10 题左右，范围：全部。

简答题强调整个学期讲解过的知识广度的理解

60 分大题(较好拿分，必拿)：

可能有的大题：

1. CPT 条件概率建模
2. PGM 的场景设计
3. elimination variation 算法
4. 贝叶斯回归和贝叶斯分类
5. 计算 VC 维，PAC 理论
6. CNN Architecture/RNN
7. Perception Training 过程
8. SVM
9. MultiArmBandit 建模和计算
10. Bayesian Network 和 MRF 学习

难点，往届反馈：

1. 考的知识点广，自己的针对性复习没抓到点
2. 时间紧，2h 题量大，往年开卷都难有限时间找到知识点，这个学期首次闭卷
3. 贝叶斯统计，概率图模型那一块计算和理解有问题

H1/Pass 的区别：
Pass：重点的知识点掌握好了，题型熟悉
H1：知识掌握全，slide 上每个角落都知道

复习方法：2 遍复习，1 遍全复习，第二遍重点复习

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PART 3 CPU 期末班针对方案

重点复习加刷题结合：

我们会把上面的分析进行知识点的分类，每个知识点讲解完后会进行大量刷题

广度复习加与课件建立记忆通道：

课堂上的复习知识点内容会与实际的官方课件 Page 建立联系，把官方课件有可能出小题的考点全面复习，并且建立记忆通道，帮助同学们在考前二轮复习快速扫盲

课程结构：

公开课：频率学派与贝叶斯学派对比，线性回归，正则化分析，逻辑回归；

课时 1： PAC 理论（计算泛化误差上界，shatter 概念及证明，Growth function 计算，如何计算并证明 VC 维）；cross validation

课时 2： SVM 算法；Kernel 的定义与证明；Perception Training；DNN 和 AutoEncoder；深度学习优化器

课时 3：CNN 和 RNN；多专家学习+Multiarmbandit；

课时 4：贝叶斯回归和贝叶斯分类；概率图的定义，应用，基于概率图的建模场景类题型及运算过程（Eliminate 算法）； CPT table 建模题型；

课时 5. Bayesian Network 学习以及 MRF 学习，样卷详解

直播答疑 2h Live 答疑

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PART 4 Statistical School of thought

1 Frequentist Statistics

- Assume fixed correct value for unknown parameter
- Maximum Likelihood Estimation (MLE):

$$\hat{\theta}(x_1, \ldots, x_n) \in \arg\max_{\theta \in \Theta} \prod_{i=1}^{n} p_\theta(x_i) \tag{1}$$

- Steps for MLE:
  1. Write the likelihood function of the dataset
  2. Take log to convert product to sum
  3. Take partial derivatives of the log-likelihood with respect to θ
  4. Set derivative to 0 and solve (or use gradient descent)

Estimator Properties

- **Bias:** $B_\theta(\hat{\theta}) = \mathbb{E}_\theta[\hat{\theta}(X_1, \ldots, X_n)] - \theta$
- **Variance:** $\mathrm{Var}_\theta(\hat{\theta}) = \mathbb{E}_\theta[(\hat{\theta} - \mathbb{E}_\theta[\hat{\theta}])^2]$
- **Consistency:** $\hat{\theta}(X_1, \ldots, X_n) \to \theta$ in probability
- **Asymptotic Efficiency:** $\mathrm{Var}_\theta(\hat{\theta}) \to$ smallest possible variance

2 Statistical Decision Theory

- Loss function measures discrepancy between prediction and truth:

$$l(a, \theta) \tag{2}$$

- Generalization error:

$$R_\theta[\delta] = \mathbb{E}_{X \sim \theta}[l(\delta(X), \theta)] \tag{3}$$

- Empirical risk:

$$\hat{R}_\theta[\delta] = \frac{1}{n} \sum_{i=1}^{n} l(\delta(x_i), \theta) \tag{4}$$

- Bias-Variance decomposition:

$$\mathbb{E}_\theta[(\theta - \hat{\theta})^2] = [B(\hat{\theta})]^2 + \mathrm{Var}_\theta(\hat{\theta}) \tag{5}$$

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3 Bayesian Statistics

- Use distributions to describe unknown parameters
- Bayes Rule:

$$P(\theta|X=x) = \frac{P(X=x|\theta)P(\theta)}{P(X=x)} \tag{6}$$

- Marginalization:

$$P(X=x) = \sum_t P(X=x, \theta=t) \tag{7}$$

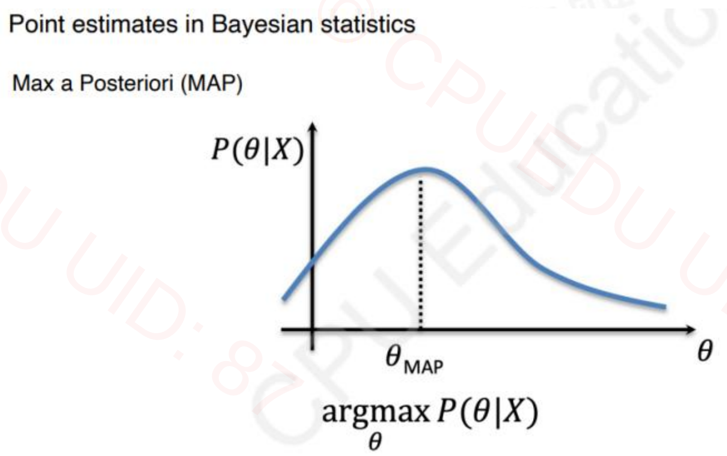
- MAP estimation:

$$\hat{\theta}_{MAP} = \arg\max_{\theta} P(\theta|X) \tag{8}$$

- Conjugate prior:

posterior $\propto$ prior $\times$ likelihood

  - If prior and posterior are in same family, prior is conjugate



习题练习

With respect to training machine learning models, how does the maximum a posteriori estimate for model weights relate to the Bayesian posterior distribution over the weights?

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Why must the evidence be computed when evaluating a Bayesian posterior, but when maximising the same posterior to find the max a posteriori (MAP) estimate, the evidence can be ignored/cancelled?

Describe how the *frequentist* and *Bayesian* approaches differ in their modelling of *unknown parameters*. [5 marks]

In words or a mathematical expression, what is the *marginal likelihood* for a *Bayesian probabilistic model*? [5 marks]

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Consider an i.i.d. sequence of random variables $X_1, X_2, \ldots$ coming from some distribution with mean θ. Consider a simple estimator $\hat{\theta}_n = \hat{\theta}(X_1, \ldots, X_n) = X_1$ of the mean. That is, use the first observation X_1 as the estimate and ignore the rest.

(a) Why is $\hat{\theta}_n$ an *unbiased estimator* of θ in general? [3 marks]

(b) Is the estimator $\hat{\theta}_n$ *consistent*? [3 marks]

(c) Explain why your answer to the previous part is correct. [6 marks]

PART 5 Linear Regression

4 Linear Regression

- From decision theory:

$$\hat{y} = \sum_{i=0}^{m} x_i w_i = x^T w \tag{9}$$

- Loss: sum of squared errors (SSE)

$$L = \sum_{i=1}^{n} \left(y_i - \sum_{j=0}^{m} x_{ij} w_j \right)^2 \tag{10}$$

- Solve:

$$\hat{w} = (X^T X)^{-1} X^T y \tag{11}$$

- Probabilistic model:

$$Y = X^T w + \varepsilon \tag{12}$$

$$\varepsilon \sim \mathcal{N}(0, \sigma^2) \tag{13}$$

$$p(y|x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(y - x^T w)^2}{2\sigma^2} \right) \tag{14}$$

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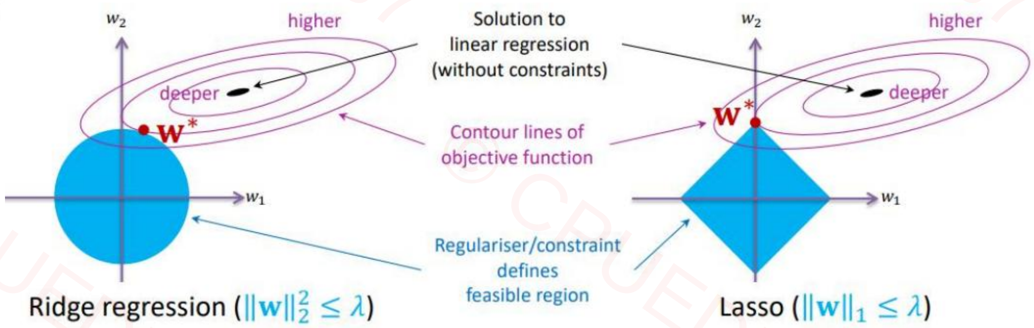
- Regularization:
  - **L2 Regularization (Ridge):**

$$L = \sum_{i=1}^{n} (y_i - x_i^T w)^2 + \lambda \|w\|_2^2 \tag{15}$$

  - **L1 Regularization (Lasso):**

$$L = \sum_{i=1}^{n} (y_i - x_i^T w)^2 + \lambda \|w\|_1 \tag{16}$$



习题练习

Show that using MLE to estimate the weight vector w (assuming variance is known) is equivalent to minimizing the sum of squared loss

Consider as an alternate to the popular square loss in supervised regression, the function $l(y;\hat{y}) = (\hat{y} - y)^5$ measuring loss between label $y \in \mathbf{R}$ and prediction $\hat{y} \in \mathbf{R}$. Is this loss a good idea or a bad idea? Why?

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In words or a mathematical expression, what quantity is minimised by *linear regression*? [5 marks]

PART 6 Logistic Regression

5 Logistic Regression

- From frequentist statistics:

$$p(y_i|x_i) = (\theta(x_i))^{y_i}(1-\theta(x_i))^{1-y_i}, \quad \theta(x_i) = \frac{1}{1+\exp(-x_i^T w)} \tag{17}$$

- Log-likelihood:

$$\log\left(\prod_{i=1}^{n} p(y_i|x_i)\right) = \sum_{i=1}^{n} \log p(y_i|x_i) \tag{18}$$

$$= \sum_{i=1}^{n} \left((y_i - 1)x_i^T w - \log(1+\exp(-x_i^T w))\right) \tag{19}$$

- From decision theory:

$$\text{Loss for } x_i : -[y_i \log(\theta(x_i)) + (1-y_i)\log(1-\theta(x_i))] \tag{20}$$

习题练习

For frequentist linear regression, we can use normal equations to solve for w and b. Can we use similar method to find the parameter values of Logistic Regression?

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Is logistic regression a linear model or non-linear model? Explain your answer.

PART 7 综合往年习题练习

21S1：

5. Consider a setting with high *uncertainty* over model parameters. Describe what effect, if any, this will have for the *maximum likelihood estimate*. [4 marks]

6. Explain why using the training likelihood, $p(\mathbf{y}|\mathbf{X}, \theta)$, for model selection can be problematic when choosing between models from different families. [4 marks]

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10. *Conjugacy* between the *likelihood* and *prior* is very important when using *Bayesian models*. However conjugacy is not critical when using the *maximum a posteriori (MAP)* estimator. Explain why this is the case. [4 marks]

22s1

(g) Why are both *maximum-likelihood estimators* and *maximum a posteriori estimators* both *asymptotically efficient*? [5 marks]

23s1

(f) Why must the *evidence* be computed when evaluating a *Bayesian posterior*, but when maximising the same posterior to find the *max a posteriori (MAP)* estimate, the *evidence* can be ignored/cancelled? [5 marks]

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23s2

(a) You have a coin that you suspect is biased. You flip it 4 times and get the sequence HHHT, where H represents *heads* and T is *tails*. What is the *maximum likelihood estimate* for the probability of obtaining heads with this coin? [5 marks]

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